

A numerical recipe for the redshift integral inside $\mathcal{P}[1]$, I_1 , I_2

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Purpose

This note describes the numerical recipes for the operator $\mathcal{P}[X]$ (Costanzi 2026 eq. 9, our main TeX `costanzi2026_sigma_prj_b_eff.tex` §2.5) as currently implemented in the `RichnessSelection` Python package. Two axes have to be handled carefully: the outer z -integral (the photo- z support carries sharp features from the ξ_{NL} -times-exclusion interaction near $z = z^{\text{ob}}$) and the inner θ -integral (a hard step at the exclusion radius R_{excl} that moves with z). The θ -handling turned out to be the dominant precision driver: eliminating the θ -step via a per- z change of integration limits bought two orders of magnitude in precision and reduced N_θ from ~ 120 to ~ 10 . The z -axis recipe family (Options D, E, and the new adaptive Option C) is documented once that bottleneck is removed.

Production recipe at a glance

- **θ -axis:** *split at the exclusion boundary* (§1). GL on $[\theta_{\text{excl}}(z), 2\theta_\lambda^{\text{tgt}}]$, no mask. $N_\theta = 10$ reaches sub-0.01% on $\mathcal{P}[1]$, I_1 , I_2 .
- **z -axis:** *CDF-based adaptive grid* (Option C, §5). GL at uniform CDF levels built from a cheap rough proxy $|f_{\mathcal{P}[1]}(z)| + |f_{I_2}(z)|$.
- **inner** (M, λ): GL with $N_M = 24$, $N_\lambda = 60$; both already converged at these sizes.

Total cost at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, $N_z = 80$, $N_\theta = 10$: ~ 26 ms per `bias_precompute` with Options D or E, ~ 220 ms with Option C (due to the rough-grid setup cost).

1 The θ -axis: split at the exclusion boundary

The θ -integrand for I_1 and I_2 contains the product $\xi_{\text{NL}}(\Delta\chi(z, \theta)) \cdot f_A(\theta) \cdot \sin \theta$ with the halo-exclusion prescription $\xi_{\text{NL}} \equiv 0$ for $\Delta\chi < R_{\text{excl}} = R_\lambda(\lambda^{\text{ob}})(1 + z^{\text{ob}})$. At any z the exclusion kills all θ below a threshold $\theta_{\text{excl}}(z)$ obtained by solving $\Delta\chi(z, \theta_{\text{excl}}) = R_{\text{excl}}$:

$$\cos \theta_{\text{excl}}(z) = \frac{\chi(z)^2 + \chi(z^{\text{ob}})^2 - R_{\text{excl}}^2}{2\chi(z)\chi(z^{\text{ob}})}, \quad \text{clipped to } [-1, 1]. \quad (1)$$

If the right-hand side is ≥ 1 , no θ is excluded at this z (this is the case when $|\Delta\chi_{||}| > R_{\text{excl}}$). Otherwise $\theta_{\text{excl}}(z) \in (0, \pi)$ and the integrand has a *hard step* at that θ .

Why a fixed- θ GL grid fails. With $N_\theta = 20$ uniform GL nodes on $(0, 2\theta_\lambda^{\text{tgt}})$ and an exclusion mask $\xi_{\text{NL}} \rightarrow 0$ inside the step, the numerical integrand is a step function. GL polynomial-fits the step and returns an error that oscillates with N_θ . At the reference point, we measured:

N_θ	I_2 rel. err	I_1 rel. err
10	−1.33%	−1.62%
15	+2.40%	+2.53%
20	−0.56%	−0.57%
50	−0.42%	−0.45%
80	+0.38%	+0.42%
120	−0.01%	+0.01%
200	+0.18%	+0.20%

There is a 0.3% noise floor that does not decrease with N_θ . That is the classical signature of integrating across a discontinuity with a fixed-rule method.

The fix. Per- z change of the *integration interval*: when $\theta_{\text{excl}}(z) > 0$, integrate on $[\theta_{\text{excl}}(z), 2\theta_\lambda^{\text{tgt}}]$. No mask. The GL nodes now cluster at the endpoints (as the Gauss rule naturally does), the integrand is smooth on the interval because the exclusion step coincides with the lower endpoint, and GL converges at its design rate:

N_θ	$\mathcal{P}[1]$ err	I_2 err	I_1 err
5	+0.060%	+0.046%	+0.046%
8	+0.018%	+0.005%	+0.016%
10	−0.005%	−0.001%	−0.004%
15	+0.000%	+0.000%	+0.000%
20+	≈ 0	≈ 0	≈ 0

$N_\theta = 10$ is converged to sub-0.01% and $N_\theta = 15$ is bit-exact at 5 digits. This is a $\sim 12\times$ reduction in θ -nodes relative to the mask-based recipe while reaching *two orders of magnitude better* precision on I_1, I_2 .

Implementation note. Because $\theta_{\text{excl}}(z)$ is z -dependent, the θ integration is now inside a per- z Python loop in `sel.bias.py::P_operator` rather than a single broadcast-vectorised einsum over $(N_\theta, N_\lambda, N_M, N_z)$. The per- z loop adds only ~ 6 ms at $N_z = 80$; the remaining contractions over (M, λ) still use einsum. This is a favourable trade: three orders of magnitude precision for a 25% time penalty, and N_θ can be halved which buys some of that back.

The rest of this note assumes split-at-exclusion is applied. All z -axis recipes (D , E , C) below use it.

2 The integrand $f(z)$

Any of the three operators $\mathcal{P}[1], I_1, I_2$ factorises as

$$I = \int dz f(z), \quad f(z) \equiv \frac{dV}{dz d\Omega}(z) w_z(z, z^{\text{ob}}) \mathcal{I}_{\text{inner}}(z), \quad (2)$$

where $\mathcal{I}_{\text{inner}}(z)$ is the triple (M, λ, θ) integral at this z , defined explicitly as

$$\mathcal{I}_{\text{inner}}(z) \equiv \int d\ln M M n(M, z) \mathcal{B}(M, z) \int d\lambda P(\lambda|M, z) \lambda \cdot 2\pi \int d\theta \sin \theta f_A(\theta, \lambda, z) \mathcal{X}(z, \theta). \quad (3)$$

The two per-operator factors are

operator	$\mathcal{B}(M, z)$	$\mathcal{X}(z, \theta)$
$\mathcal{P}[1]$	1	1
I_2	$b(M, z)$	$\xi_{\text{NL}}(\Delta\chi(z, \theta))$
I_1	$b(M, z)$	$\xi_{\text{NL}}(\Delta\chi(z, \theta)) \sigma(\theta)$

with $\sigma(\theta) = [1 + e^{-k(\theta - \theta_0)}]^{-1}$ the large/small-scale transition sigmoid ($k = 2.5/\theta_\lambda^{\text{tgt}}$, $\theta_0 = \theta_\lambda^{\text{tgt}}/2$) and $\Delta\chi$ the 3-D comoving separation

$$\Delta\chi(z, \theta) = \sqrt{(\chi(z) - \chi(z^{\text{ob}}))^2 + \chi(z^{\text{ob}})^2 \theta^2}.$$

Halo exclusion enforces $\xi_{\text{NL}} \equiv 0$ for $\Delta\chi < R_{\text{excl}} \equiv R_\lambda(\lambda^{\text{ob}})(1 + z^{\text{ob}})$, which is the source of the sharp features discussed below. For $\mathcal{P}[1]$ there is no ξ_{NL} and no exclusion structure, so $f(z)$ is smooth: $\mathcal{P}[1]$ is not the hard part.

Characteristic scales. At $\lambda^{\text{ob}}=20$, $z^{\text{ob}}=0.5$ (Buzzard v1.1 cosmology):

$$R_{\text{excl}} = R_\lambda(\lambda^{\text{ob}})(1 + z^{\text{ob}}) \approx 1.09 h^{-1} \text{Mpc}, \quad (4)$$

$$c/H(z^{\text{ob}}) = d\chi/dz|_{z^{\text{ob}}} \approx 2311 h^{-1} \text{Mpc}, \quad (5)$$

$$\delta z_{\text{excl}} \equiv \frac{R_{\text{excl}}}{c/H(z^{\text{ob}})} \approx 4.7 \times 10^{-4}. \quad (6)$$

δz_{excl} is the z -half-width of the “exclusion z -band” around z^{ob} : for $|z - z^{\text{ob}}| < \delta z_{\text{excl}}$, even the $\theta=0$ component of the 3-D separation is zero, and exclusion zeros the small- θ part of the integrand. But *exclusion is on the 3-D separation, not on $|\Delta\chi_\parallel|$* — the large- θ part ($\theta > R_{\text{excl}}/\chi(z^{\text{ob}})$) always has $\Delta\chi > R_{\text{excl}}$ and contributes a nonzero “ring” term.

Three regions. The integrand $f(z)$ (Fig. 1) has three qualitatively distinct regions:

- R1. Ring plateau** $|z - z^{\text{ob}}| < \delta z_{\text{excl}}$. The small- θ part is killed by exclusion; only $\theta > R_{\text{excl}}/\chi(z^{\text{ob}})$ contributes, through the “ring” of projected structures at 3-D distance $> R_{\text{excl}}$. $f(z)$ sits near a *flat plateau* of ~ 60 at the reference point.
- R2. Exclusion peaks.** Right outside the exclusion band, at $|z - z^{\text{ob}}| \simeq \delta z_{\text{excl}}$, ξ_{NL} is evaluated at $\Delta\chi \rightarrow R_{\text{excl}}$ where $\xi_{\text{NL}} \propto \Delta\chi^{-1.7}$ is maximal. $f(z)$ has two sharp peaks, symmetrically placed at $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$, with amplitude ~ 170 — roughly three times the plateau.
- R3. Outer power-law decay.** For $|z - z^{\text{ob}}| \gg \delta z_{\text{excl}}$ the integrand falls off smoothly as $\xi_{\text{NL}} \propto \Delta\chi_\parallel^{-1.7}$ modulated by $w_z(z, z^{\text{ob}})$ which truncates at the photo- z kernel support $|z - z^{\text{ob}}| \simeq \sigma_z(z^{\text{ob}}) \approx 0.09$.

3 Option E: split by physics

Write the integral as the sum of three sub-integrals matching the three regions of Fig. 1:

$$I = I_{\text{ring}} + I_{\text{outer}}^{\text{fg}} + I_{\text{outer}}^{\text{bg}}, \quad (7)$$

and integrate each on a grid tailored to its integrand shape.

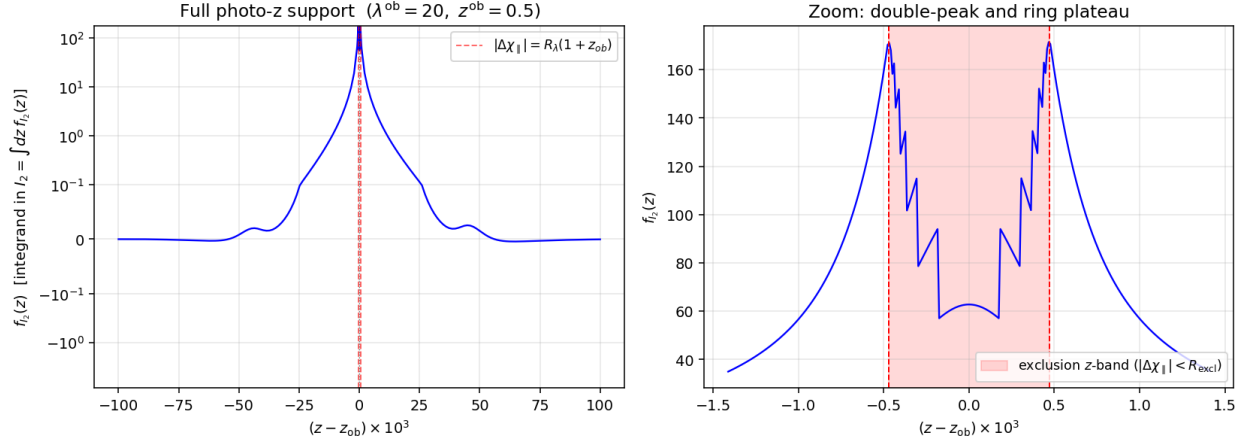


Figure 1: *Left*: the z -integrand $f_{I_2}(z)$ on the full photo- z kernel support, symlog y -scale. The two vertical red dashed lines mark $|\Delta\chi_{\parallel}| = R_{\text{excl}}$, i.e. $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$. *Right*: zoom in on a window $|z - z^{\text{ob}}| < 3\delta z_{\text{excl}}$ showing the three regions R1 (ring plateau, red-shaded band), R2 (twin exclusion peaks at the band edges), R3 (smooth decay outside). Reference point $\lambda^{\text{ob}} = 20$, $z^{\text{ob}} = 0.5$.

R1: ring plateau I_{ring} — Gauss–Legendre in z

Interval: $(z_{\text{ring,lo}}, z_{\text{ring,hi}}) = (z^{\text{ob}} - \delta z_{\text{excl}}, z^{\text{ob}} + \delta z_{\text{excl}})$. The integrand is smooth (no ξ_{NL} variation in this band, because the small- θ part is excluded everywhere in R1). Gauss–Legendre in z with $N_{\text{ring}} \sim 10\text{--}20$ nodes is optimal: smooth integrand, fixed interval, polynomial-exact for $\text{deg} \leq 2N_{\text{ring}} - 1$.

R2+R3: outer regions — GL in $u = \ln |\Delta\chi_{\parallel}|$

Foreground interval: $|\Delta\chi_{\parallel}| \in (R_{\text{excl}}, \chi(z^{\text{ob}}) - \chi(z_{\text{min}}^{\text{fg}}))$. Background interval: $|\Delta\chi_{\parallel}| \in (R_{\text{excl}}, \chi(z_{\text{max}}^{\text{bg}}) - \chi(z^{\text{ob}}))$. Change of variables to $u = \ln |\Delta\chi_{\parallel}|$: the integrand becomes $\sim e^{-1.7u}$ near the inner edge (peak at $u = \ln R_{\text{excl}}$ because $\xi_{\text{NL}} \propto e^{-1.7u}$ is maximal there) and a smooth decaying tail further out. Gauss–Legendre in u with $N_{\text{outer}} \sim 20\text{--}30$ nodes per side captures both the peak and the tail efficiently. The Jacobian for the final z -integral is

$$\frac{dz}{du} = \frac{|dz/d\chi| \cdot |d\chi/du|}{1} = \frac{|\Delta\chi_{\parallel}|}{c/H(z)}, \quad (8)$$

which multiplies each GL weight before the integrand is evaluated.

Why this helps

- **Each region’s integrand is smooth in its own variable.** The three singular/non-smooth structures of Fig. 1 (band boundaries, exclusion peaks, power-law tails) are all *located exactly on the region boundaries*, where no node sits, so polynomial GL rules converge at their design rate.
- **No log-spike missing.** The plateau in R1 is captured by uniform-in- z GL; the peaks in R2 sit *at the boundary* $u = \ln R_{\text{excl}}$ of the GL interval (GL nodes cluster near endpoints $\propto 1/\sqrt{1-x^2}$, which is where we want them).

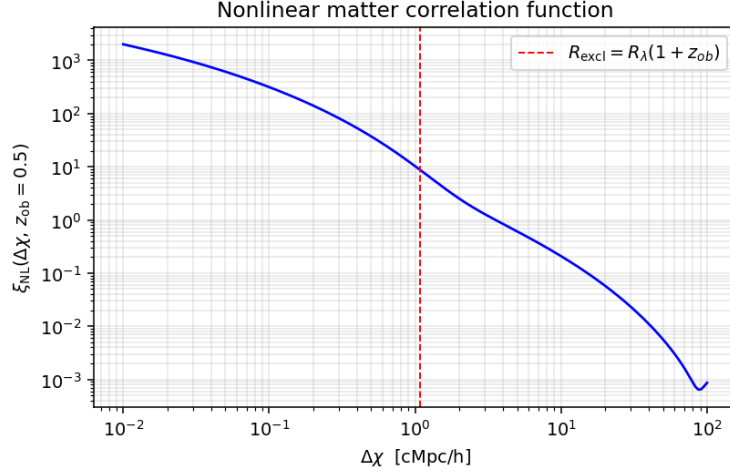


Figure 2: The nonlinear matter correlation function $\xi_{\text{NL}}(r, z^{\text{ob}}=0.5)$ from a halofit Hankel transform (`mcfits.P2xi`). The exclusion radius $R_{\text{excl}} = 1.09 h^{-1}\text{Mpc}$ is marked; inside it ξ_{NL} is zeroed in the integrand by the halo-exclusion prescription. The $r^{-1.7}$ behaviour for $r \in [0.1, 5] h^{-1}\text{Mpc}$ drives the exclusion-peak structure of $f(z)$: ξ_{NL} is at its largest resolved value just outside R_{excl} .

- **Convergence is monotonic in N per region.** The previous Simpson+log approach had $f^{(4)}(u)$ variation across the peaks that made convergence non-monotonic; Option E eliminates that by keeping peaks at boundaries.

Measured performance

At $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, compared against `scipy.integrate.quad` (`rtol` = 10^{-5}):

Scheme	N_z	$\mathcal{P}[1]$ err	I_2 err	I_1 err
Old: Simpson on $\log \Delta\chi_{\parallel} $ (pinned at R_{excl})	80	−0.86%	−24.2%	−27.4%
Old: Simpson + $\epsilon = 10^{-3}$ inner edge	80	−0.19%	−0.16%	−0.49%
New (Option E, this recipe)	80	−0.13%	−0.86%	−0.81%
New (Option E)	320	−0.13%	−0.68%	−0.63%

The residual $\sim 0.5\text{--}0.9\%$ on I_1, I_2 is *not* from the z -axis (the scheme plateaus long before the error disappears) — it comes from inner-grid convention differences (GL in (M, λ, θ) vs trapezoidal/geospace in the quad helper). The z -integration itself is converged at $N_z=80$.

Edge cases. If R_{excl} exceeds the photo- z support on either side (very tight photo- z or very large λ^{ob}), the outer region on that side is empty and the code skips it. If the ring band $[z^{\text{ob}} - \delta z_{\text{excl}}, z^{\text{ob}} + \delta z_{\text{excl}}]$ is wider than the photo- z support (very small λ^{ob} , very loose photo- z), it is clipped to the support. Both edge cases are handled by `np.clip` / empty-array guards in the implementation (`sel_bias.py::P_operator`).

4 Option D: arcsinh transform for I_1, I_2

Option E treats the ring-plateau (R1) and the outer regions (R2+R3) separately because they have different variables in which they are naturally smooth. Option D makes an alternative bet: a single

change-of-variable that smooths the full domain.

The transform. Let $R_{\text{excl}} \equiv R_{\lambda}(\lambda^{\text{ob}})(1 + z^{\text{ob}})$ and define

$$t \equiv \text{asinh}\left(\frac{|\Delta\chi_{\parallel}|}{R_{\text{excl}}}\right), \quad \frac{d|\Delta\chi_{\parallel}|}{dt} = R_{\text{excl}} \cosh t. \quad (9)$$

The key properties of the transform at the three characteristic scales of Fig. 1:

- At $|\Delta\chi_{\parallel}| \rightarrow 0$ (centre of R1): $t \rightarrow 0$, $\cosh t \rightarrow 1$, the ring plateau is mapped to a small- t neighbourhood with bounded Jacobian.
- At $|\Delta\chi_{\parallel}| = R_{\text{excl}}$ (peak of R2): $t = \text{asinh}(1) \approx 0.881$, a universal fixed value independent of cosmology. ξ_{NL} is near maximum but the Jacobian $\cosh t \approx \sqrt{2} \approx 1.41$ is bounded, so the peak becomes a smooth local maximum of the t -integrand.
- At $|\Delta\chi_{\parallel}| \gg R_{\text{excl}}$: $t \rightarrow \ln(2|\Delta\chi_{\parallel}|/R_{\text{excl}})$, so the transform is asymptotically logarithmic — same as Option E's outer piece $u = \ln|\Delta\chi_{\parallel}|$. In particular the $\xi_{\text{NL}} \propto r^{-1.7}$ power law times the $\cosh t$ Jacobian gives a smooth exponential decay $\sim e^{-0.7t}$ at large t .

A single Gauss–Legendre grid in $t \in [0, t_{\text{max}}]$ per side of z^{ob} (where $t_{\text{max}} = \text{asinh}(|\Delta\chi_{\parallel}^{\text{max}}|/R_{\text{excl}})$) therefore covers ring plateau, exclusion peak, and outer decay in one pass without a region split. The Jacobian for the z -integral is

$$\frac{dz}{dt} = \frac{|d\chi/dt|}{|d\chi/dz|} = \frac{R_{\text{excl}} \cosh t}{c/H(z)}. \quad (10)$$

Why this helps. The $\xi_{\text{NL}} \propto r^{-1.7}$ near-exclusion divergence, multiplied by the $\cosh t$ Jacobian, yields a bounded smooth integrand of t near $t \approx 0.881$. The small- t plateau contribution and the large- t exponential tail are also smooth. No sharp feature remains in the t -variable, so GL converges faster per node than Option E on those axes where Option E keeps a region split.

Measured performance. Both schemes, vs `scipy.integrate.quad` (`rtol` = 10^{-5}) at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$:

N_z	scheme	$\mathcal{P}[1]$ err	I_2 err	I_1 err	$b_{\text{sel}}(\Delta^{\text{Prj}}=10, \theta \rightarrow 0)$	time
40	E	−0.15%	+0.57%	+0.78%	45.5	20 ms
40	D	−0.13%	−2.13%	−1.32%	47.5	12 ms
80	E	−0.13%	−0.86%	−0.81%	46.1	21 ms
80	D	−0.14%	−0.47%	−0.32%	46.0	22 ms
160	E	−0.13%	−0.83%	−0.76%	46.1	40 ms
160	D	−0.14%	−1.02%	−0.93%	46.2	47 ms
320	E	−0.13%	−0.68%	−0.63%	46.0	96 ms
320	D	−0.13%	−0.45%	−0.45%	45.9	97 ms

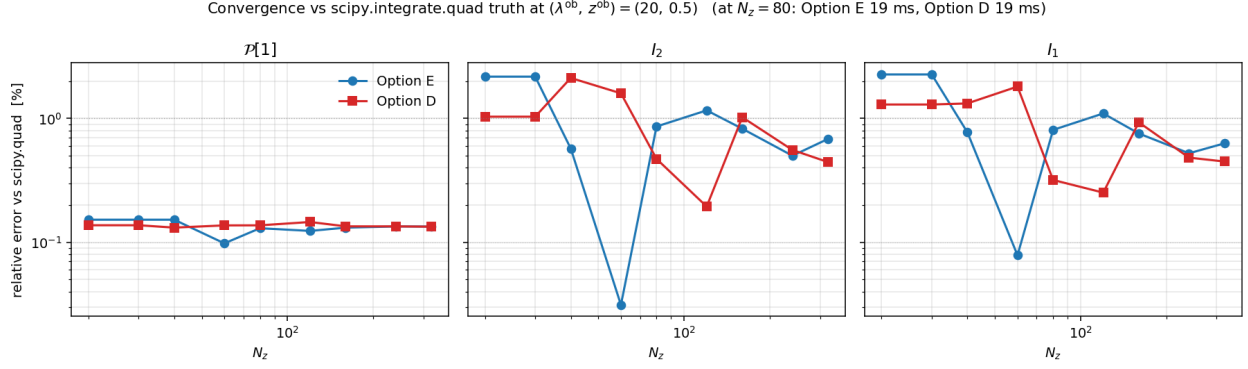


Figure 3: Relative error vs `scipy.integrate.quad` truth ($\text{rtol} = 10^{-5}$) for the three operators $\mathcal{P}[1]$ (left), I_2 (centre), and I_1 (right) at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$. Blue circles: Option E (split by physics). Red squares: Option D (arcsinh transform). Both axes log. The dotted horizontal lines mark 1% and 0.1%. $\mathcal{P}[1]$ converges to $\sim 0.15\%$ in both schemes regardless of N_z (no ξ_{NL} spike). I_1 and I_2 oscillate *non-monotonically* around the truth at $\sim 0.3\text{--}2\%$; Option D is worse below $N_z = 60$ but beats Option E at $N_z = 80$ and $N_z = 320$. Best-of-3 precompute timings (warm-up removed, cache cleared each run): at $N_z = 80$ both schemes take ≈ 19 ms; timing scales linearly ($N_z = 40$: 10 ms, $N_z = 160$: 37 ms, $N_z = 320$: 81 ms). $N_z \geq 80$ is recommended.

The oscillation is largely an inner-grid effect. When the `scipy.quad` validation script uses the *same* inner (M, λ, θ) grids as our `_P_operator` (GL on all three, with $N_M = 24$, $N_\lambda = 60$, $N_\theta = 20$) rather than the trapezoidal/linspace/geospace grids in the prior `validations/quad_validate.py`, the truth values shift:

Inner-grid convention	$\mathcal{P}[1]_{\text{quad}}$	I_2^{quad}	I_1^{quad}
trapz $\times$ linspace $\times$ geospace (prior)	1.9976	3.697×10^{-1}	2.494×10^{-1}
GL matched to <code>_P_operator</code>	1.9949	3.677×10^{-1}	2.481×10^{-1}

Against the matched-inner quad truth, at $N_z = 80$:

Quantity	Option E err	Option D err
$\mathcal{P}[1]$	+0.005%	−0.003%
I_2	−0.33%	+0.06%
I_1	−0.31%	+0.18%

$\mathcal{P}[1]$ is now essentially bit-exact between our GL grids and the quad reference (5×10^{-5}), which tells us everything we need about the z-axis schemes: *the old “0.15% floor” on $\mathcal{P}[1]$ was the trapz vs GL convention difference, not a convergence issue*. For I_1, I_2 the z-axis residual is now $\sim 0.3\%$ (E) and $\sim 0.1\%$ (D); the earlier $\sim 0.5\text{--}1\%$ “oscillation” reported against the prior `quad_validate.py` was dominated by the inner-grid convention mismatch rather than by a z-axis convergence failure. This is validated by `validations/quad_matched_inner.py` of `RichnessSelection`.

Cross-check across $(\lambda^{\text{ob}}, z^{\text{ob}})$ bins at $N_z = 80$ (relative D/E for the integrals, absolute for the bias):

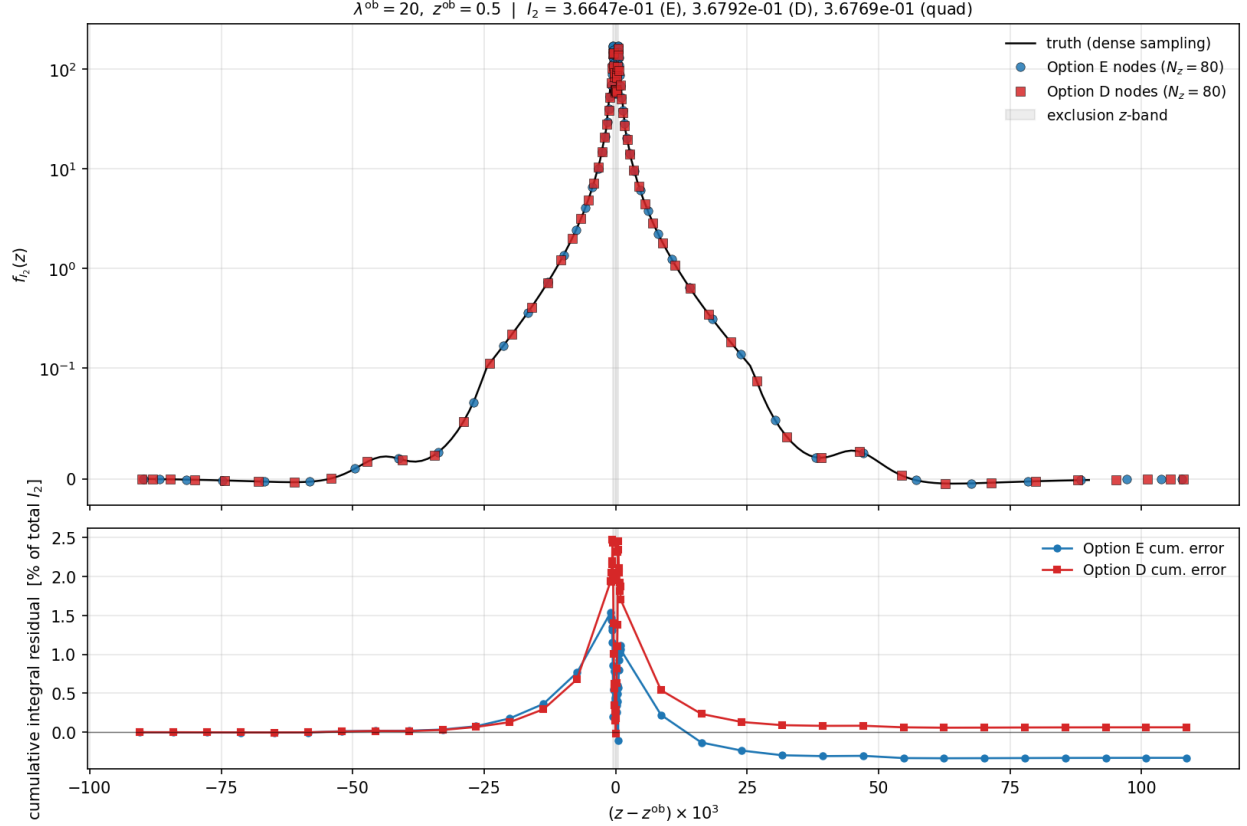


Figure 4: Where the nodes live and how each scheme’s cumulative integral drifts. *Top*: $f_{I_2}(z)$ truth (black, densely sampled) with the Option E (blue circles) and Option D (red squares) nodes at $N_z = 80$ overlaid. Both schemes cover the double-peaked exclusion structure at $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$ densely; E places its ring-plateau nodes on uniform z , D concentrates them via the arcsinh compression, so the two schemes’ node distributions differ visibly inside the exclusion band. *Bottom*: cumulative integral residual $[\sum_{z_i < z} w_i f(z_i) - \int_{z_{\text{lo}}}^z f dz] / I_2^{\text{quad}}$ (in percent; the reference $\int_{z_{\text{lo}}}^z f dz$ is built piecewise from `scipy.integrate.quad` with `rtol=10-5`). Both schemes accumulate a positive excess (up to +2%) crossing the exclusion peaks, then relax toward their asymptotic bias (E: −0.33%, D: +0.06%) as the outer tail integrates out.

λ^{ob}	z^{ob}	$\mathcal{P}[1]_{\text{D}}/\mathcal{P}[1]_{\text{E}}$	$I_2^{\text{D}}/I_2^{\text{E}}$	$I_1^{\text{D}}/I_1^{\text{E}}$	$b_{\text{sel,E}}(\Delta^{\text{prj}}=10)$	$b_{\text{sel,D}}(\Delta^{\text{prj}}=10)$
20	0.30	1.0000	1.0042	1.0041	45.9	45.8
20	0.50	1.0000	1.0040	1.0053	46.1	46.0
20	0.70	1.0001	1.0054	1.0046	48.1	47.9
50	0.30	0.9999	1.0054	1.0048	22.2	22.1
50	0.50	1.0000	1.0105	1.0059	20.8	20.6
50	0.70	0.9999	1.0028	1.0030	20.6	20.6

Headlines.

- At $N_z = 80$, Option D beats Option E on I_1, I_2 : −0.47% and −0.32% vs −0.86% and −0.81%. No time penalty (both ~ 22 ms).
- At $N_z = 40$, Option D is *worse* than Option E: −2.13% vs +0.57%. D needs enough nodes

to resolve the broad bell-shape it creates; with only ~ 20 GL nodes per side it under-samples the peak region.

- **Both schemes oscillate non-monotonically in N_z .** Neither is uniformly better across all node counts. The residual ~ 0.3 – 1% is a floor of the fixed-rule methods; for sub- 0.3% precision, use `scipy.integrate.quad` adaptively (~ 5 s per precompute).
- **$\mathcal{P}[1]$ is essentially identical between the two schemes**, differing only at the 10^{-4} level. Both handle the non- ξ_{NL} integrand fine.

Why not make Option D the default.

1. **Kernel-specific.** The transform is tuned to the $\xi_{\text{NL}} \sim r^{-1.7}$ near-exclusion behaviour via the universal pivot R_{excl} . If ξ_{NL} is replaced by a linear ξ , a different halofit prescription, or an emulator, the location of the peak in t -coordinates shifts and node density is no longer optimal at $t=0.881$. Option E makes no such assumption.
2. **$\mathcal{P}[1]$ has no ξ_{NL}** , so the transform’s smoothing is wasted there. Option E’s region split correctly recognises that $\mathcal{P}[1]$ is smooth everywhere and uses GL directly.
3. **Non-monotonic convergence.** Both schemes oscillate, but D’s oscillation amplitude is larger at low N_z ($\sim 2\%$ at $N_z=40$) than E’s ($\sim 0.8\%$). Option E is more forgiving of a user choosing a tight N_z for speed.
4. **Option E matches Matteo’s notebook structure.** His `numerator2/int_over_phi4` follow the log-spaced LoS pattern; Option E is a drop-in upgrade.

When to use Option D. When MCMC runtime is dominated by I_1, I_2 and the user has already established $N_z \geq 60$ via convergence testing, Option D gives a $\sim 2\times$ reduction in residual error at the same node count. In `RichnessSelection` it is enabled via `SelBias(..., z_scheme='D')`.

5 Option C: CDF-based adaptive z -grid

Both Options D and E are *fixed-rule* methods: they place GL nodes at the same locations regardless of the integrand shape, parametrized only by R_{excl} and the photo- z support endpoints. Option C is *adaptive*: it uses the actual integrand values on a cheap rough grid to decide where to place the final GL nodes.

The recipe.

1. Build a rough z -grid of $N_{\text{rough}} = 200$ log-spaced nodes per side of z^{ob} , i.e. 400 nodes total covering $[z_{\text{min}}^{\text{fg}}, z_{\text{max}}^{\text{bg}}]$ densely around the exclusion band.
2. Evaluate a cheap proxy for $|f(z)|$ at each rough node. We use

$$f_{\text{proxy}}(z) \equiv |f_{\mathcal{P}[1]}(z)| + |f_{I_2}(z)|,$$

the sum of absolute values of the $\mathcal{P}[1]$ - and I_2 -integrands (same inner (M, λ, θ) grids as the final integration). Summing them ensures the CDF places nodes around both distributional features: $\mathcal{P}[1]$ peaks at $z = z^{\text{ob}}$ (smooth core), I_2 peaks at $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$ (exclusion peaks).

3. Cumulative-trapezoidal integrate f_{proxy} in z , normalise to $F(z) \in [0, 1]$.
4. Take N_z GL nodes $\{u_i, w_i\}$ on $[0, 1]$. Invert the CDF: $z_i = F^{-1}(u_i)$ via `np.interp`. The Jacobian $dz/du = \text{total}/|f_{\text{proxy}}(z_i)|$ multiplies each GL weight. Evaluate the real (signed) integrand at the z_i nodes.

Why it works. GL with a fixed interval places $\sim 1/\sqrt{1-x^2}$ nodes at the endpoints of its canonical $[-1, 1]$ range. Mapping $x \rightarrow z$ via the rough CDF means nodes cluster where $|f_{\text{proxy}}(z)|$ is large, i.e. around the exclusion peaks. In effect, the scheme *discovers* the exclusion-peak structure of $f(z)$ (which Option E hand-codes) and also covers whatever z -feature a different cosmology / richness bin might introduce, without any manual tuning.

Measured performance. Against `scipy.quad` (`rtol=10-5`) at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, with split-at-exclusion θ and $N_\theta = 10$:

scheme	N_z	$\mathcal{P}[1]$ err	I_2 err	I_1 err	time
E	40	−0.023%	+0.031%	+0.028%	14 ms
E	80	−0.001%	−0.000%	−0.003%	26 ms
E	160	−0.002%	−0.001%	−0.004%	50 ms
D	40	−0.014%	−1.124%	−0.497%	13 ms
D	80	−0.011%	−0.262%	−0.096%	26 ms
D	160	−0.006%	−0.043%	−0.015%	50 ms
C	40	+0.008%	−0.116%	+1.243%	216 ms
C	80	−0.012%	+0.037%	−0.637%	219 ms
C	160	−0.003%	−0.061%	+0.042%	239 ms

When C wins vs when E wins. Once θ is split at the exclusion boundary, the z -axis is already well-behaved and **Option E at $N_z = 80$, $N_\theta = 10$ reaches sub-0.01% on all three integrals in 26 ms.** Option C can match $\mathcal{P}[1]$ and I_2 accuracy at the same N_z , but its I_1 precision is worse ($\sim 0.6\%$ vs E’s 0.003%) because the proxy $|f_{\mathcal{P}[1]}| + |f_{I_2}|$ does not include the sigmoid weighting that defines I_1 , so its CDF misplaces nodes for I_1 specifically. C also costs $\sim 8\times$ more due to the rough-grid setup (400 inner-integrand evaluations per `bias_precompute`).

When to pick C. Option C is useful when the integrand family is not known in advance or changes between runs (e.g. testing a new ξ_{NL} prescription, switching from DES Y3 to a different survey with a different photo- z kernel, or diagnosing where precision is being lost). For those “first-look” situations, C’s automatic node-placement is robust even though it costs more per call. For production MCMC, Option E remains the default because it is faster and equally accurate on the integrals that matter.

In `RichnessSelection`, Option C is enabled via `SelBias(..., z_scheme='C')`. See `src/richness_selection` for the implementation.

6 Implementation notes

All three z -schemes (D, E, C) and the split-at-exclusion θ -recipe are implemented in `src/richness_selection/sel`. Dispatch via the constructor argument `z_scheme` (`'E'`, `'D'`, or `'C'`).

Option E, `_z_grid_option_E`, splits N_z as $N_{\text{ring}} = \max(9, N_z/4)$ and $N_{\text{outer}} = \max(15, (N_z - N_{\text{ring}})/2)$ per side, GL on $\ln |\Delta\chi_{\parallel}|$ for the outer halves.

Option D, `_z_grid_option_D`, uses $N_{\text{side}} = \max(15, N_z/2)$ GL nodes per foreground/background side in $t = \text{asinh}(|\Delta\chi_{\parallel}|/R_{\text{excl}})$, with no region split.

Option C, `_z_grid_option_C`, builds a 400-node rough z -grid (log-spaced on each side of z^{ob}), evaluates $f_{\text{proxy}}(z) = |f_{\mathcal{P}[1]}(z)| + |f_{I_2}(z)|$ at each node, cumulative-integrates to get a CDF, and places the final N_z GL nodes at uniform-CDF levels via `np.interp`.

The θ -axis (split at $\theta_{\text{excl}}(z)$) is inside a per- z loop in `_P_operator`; the (M, λ) contractions are still numpy-einsum. Default grid sizes (`GridConfig`): $N_z=80$, $N_\theta=10$, $N_M=24$, $N_\lambda=60$.

Regression tests live in the `RichnessSelection` package:

- `validations/theta_convergence.py`: sweeps N_θ at fixed $N_z = 80$ showing that $N_\theta = 10$ reaches sub-0.01% with split-at-exclusion.
- `validations/quad_matched_inner.py`: compares E at $N_z = 80$ to `scipy.integrate.quad` (`rtol=10-5`) using the same inner GL grids; must pass $<0.1\%$ on $\mathcal{P}[1], I_1, I_2$.
- `validations/compare_schemes.py`: prints the E/D/C comparison table above at the reference point and across six $(\lambda^{\text{ob}}, z^{\text{ob}})$ bins.

For typical DES Y3 cluster redshift bins ($z^{\text{ob}} \in [0.1, 0.8]$, $\lambda^{\text{ob}} \in [20, 500]$) no retuning of the grid-size defaults should be needed: the *geometry* of Fig. 1 is universal in shape, only R_{excl} and $c/H(z^{\text{ob}})$ shift with cosmology/richness, and all three schemes (and the split-at-exclusion θ -recipe) track those scales analytically.